

8th-Grade

TOPICS

1. Real Numbers
    - Reviews
    - Powers and Radicals
2. Algebraic Expressions
    - Definition of Algebraic Expressions
    - Identities and Factorization
    - Algebraic Ratios
3. Sequences and Functions
    - Sequences
    - Functions
    - Special Functions
4. Equations and Inequalities
    - Equations
    - System of Equations
    - Inequalities
5. Quadratic Equations
6. Probability and Statistics
    - Probability
    - Statistics
7. Mathematical Logic
8. Equations and Inequalities
    - Basic Geometric Concepts
    - Angle and Similarity in Triangles
    - Right Triangles and Trigonometry
    - Quadrilaterals

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1. Real Numbers
   - Reviews

1.REAL NUMBERS

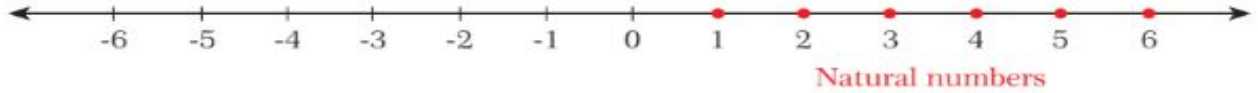
1.1 REVIEW

A. SET OF REAL NUMBERS

1. Understanding Real Numbers

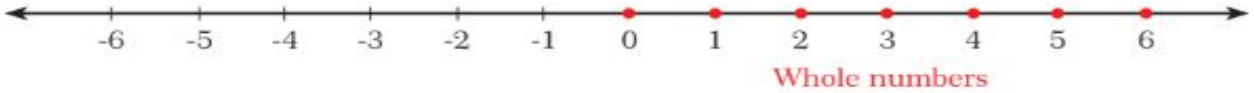
The set of natural numbers is denoted by N.

$N = \{1, 2, 3, 4, 5, ...\}$



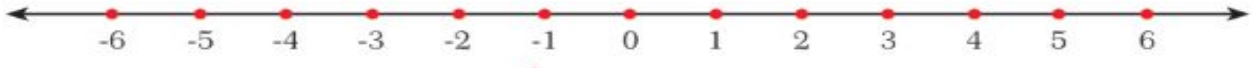
The set of whole numbers is the set of natural numbers together with zero. It is denoted by W.

$W = \{0, 1, 2, 3, 4, 5, ...\}$



The set of integers is the set of natural numbers, together with zero and the negatives of the natural numbers. It is denoted by Z.

$Z = \{..., -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, ...\}$



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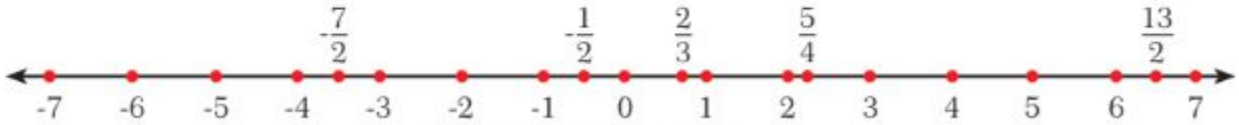
1.1 REVIEW

A. SET OF REAL NUMBERS

1. Understanding Real Numbers

The set of rational numbers is the set of numbers that can be written as the quotient of two integers. It is denoted by Q.

$$Q = \{ \frac{a}{b} \mid a, b \in \mathbb{Z} \text{ and } b \neq 0 \}$$



Some rational numbers

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1.1 REVIEW

A. SET OF REAL NUMBERS

1. Understanding Real Numbers

Definition

A number whose decimal form does not repeat or terminate is called an irrational number. The set of **irrational** numbers is denoted by Q' or I.

Definition

The union of the set of rational numbers and the set of irrational numbers forms the set of all decimals. This union is called the set of **real numbers**.

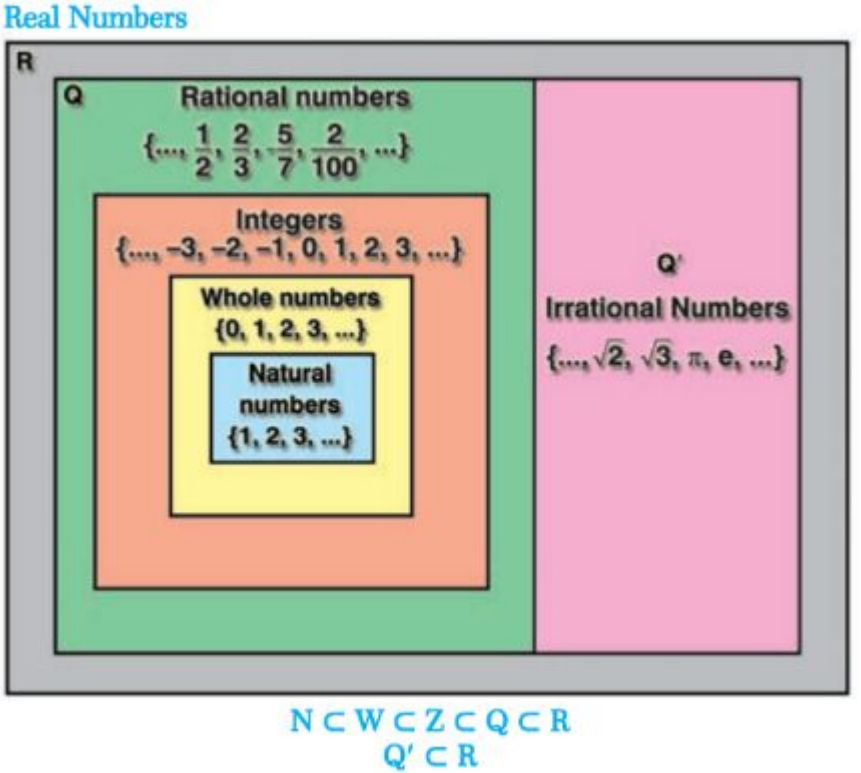
The set of real numbers is denoted by R.

$$R = Q \cup Q'$$

1.1 REVIEW

A. SET OF REAL NUMBERS

1. Understanding Real Numbers



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